

Lab 9: BJTs (2)

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	Student Name
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab is to study the behavior of the current gain β of a bipolar junction transistor (BJT). A 2N2222A NPN transistor was biased using a collector resistor R_C and an emitter resistor R_E . The base voltage V_B was varied from 2 V to 8 V, and the corresponding base current, collector voltage, and emitter voltage were measured.

For a BJT, the common-emitter current gain and common-base current gain are defined as

$$\beta = \frac{I_C}{I_B}$$

and

$$\alpha = \frac{I_C}{I_E}$$

In the forward-active region, β is relatively large and the collector current is controlled mainly by the base current. In saturation, V_{CE} becomes very small, and the collector current is limited mainly by the external collector resistor and the supply voltage. Therefore, the forced current gain $\beta = I_C/I_B$ decreases sharply at high V_B .

2) Equipment Used

Table 1: Equipment used in Lab 9

Item	Quantity
2N2222A NPN transistor	1
Resistor, nominal 4.7 k Ω	1
Resistor, nominal 3.3 k Ω	1
DC Power Supply	2
Digital Multimeter	1
Breadboard and wires	1 set

3) Action 1: Node Voltages and Region of Operation

Case 1: $V_B = 4\text{ V}$

Assuming forward-active operation,

$$V_E = V_B - V_{BE} = 4.0 - 0.7 = 3.3\text{ V}$$

Then

$$I_E = \frac{V_E}{R_E} = \frac{3.3}{3.3\text{ k}\Omega} = 1.00\text{ mA}$$

Since $\alpha = 1$,

$$I_C \approx I_E = 1.00\text{ mA}$$

Thus,

$$V_C = V_{CC} - I_C R_C = 10 - (1.00\text{ mA})(4.7\text{ k}\Omega) = 5.30\text{ V}$$

The collector-emitter voltage is

$$V_{CE} = V_C - V_E = 5.30 - 3.30 = 2.00\text{ V}$$

Since

$$V_{CE} = 2.00\text{ V} > V_{CE,sat} = 0.1\text{ V}$$

the transistor is in the forward-active region.

Table 2: Measured node voltages and collector current for $V_B = 4\text{ V}$

Set V_B	Measured V_B [V]	V_C [V]	V_E [V]	V_{CE} [V]	I_C [mA]
4 V	3.998	5.153	3.388	1.765	1.032

Case 2: $V_B = 6\text{ V}$

If forward-active operation is assumed,

$$V_E = V_B - V_{BE} = 6.0 - 0.7 = 5.3\text{ V}$$

Then

$$I_E = \frac{5.3}{3.3\text{ k}\Omega} = 1.61\text{ mA}$$

and

$$I_C \approx 1.61\text{ mA}$$

This would give

$$V_C = 10 - (1.61 \text{ mA})(4.7 \text{ k}\Omega) = 2.44 \text{ V}$$

Then

$$V_{CE} = V_C - V_E = 2.44 - 5.30 = -2.86 \text{ V}$$

This result is impossible for normal forward-active operation. Therefore, the forward-active assumption fails, and the transistor is in saturation.

In saturation,

$$V_{CE} \approx V_{CE,sat} = 0.1 \text{ V}$$

Therefore,

$$V_C \approx V_E + V_{CE,sat} = 5.3 + 0.1 = 5.4 \text{ V}$$

The collector current is approximately

$$I_C = \frac{V_{CC} - V_C}{R_C} = \frac{10 - 5.4}{4.7 \text{ k}\Omega} = 0.979 \text{ mA}$$

Table 3: Measured node voltages and collector current for $V_B = 6 \text{ V}$

Set V_B	Measured V_B [V]	V_C [V]	V_E [V]	V_{CE} [V]	I_C [mA]
6 V	5.999	5.315	5.312	0.003	0.997

4) Action 2: Measurement of I_B , V_C , and V_E

Measured Resistors

Table 4: Measured resistor values

Parameter	Measured Value
R_C	4.690 k Ω
R_E	3.269 k Ω

Data and Calculation

Table 5: Measured and calculated data for Action 2

V_B	V_C [V]	V_E [V]	I_B [μ A]	V_{CE} [V]	I_C [mA]	I_E [mA]	β	α
2 V	7.967	1.419	1.573	6.548	0.433	0.434	275.57	0.999
2.5 V	7.262	1.912	2.130	5.350	0.584	0.585	274.08	0.998
3 V	6.561	2.403	2.692	4.158	0.733	0.735	272.39	0.998
3.5 V	5.852	2.899	3.269	2.953	0.884	0.887	270.55	0.997
4 V	5.145	3.394	3.858	1.751	1.035	1.038	268.32	0.997
4.3 V	4.719	3.693	4.224	1.026	1.126	1.130	266.57	0.997
4.4 V	4.578	3.791	4.348	0.787	1.156	1.160	265.89	0.997
4.5 V	4.439	3.890	4.475	0.549	1.186	1.190	264.96	0.996
4.6 V	4.297	3.989	4.605	0.308	1.216	1.220	264.06	0.997
4.7 V	4.173	4.087	8.132	0.086	1.242	1.250	152.78	0.994
4.8 V	4.205	4.177	42.501	0.028	1.236	1.278	29.07	0.967
4.9 V	4.283	4.267	86.622	0.016	1.219	1.305	14.07	0.934
5 V	4.369	4.359	133.022	0.010	1.201	1.333	9.03	0.900
5.1 V	4.458	4.449	179.553	0.009	1.182	1.361	6.58	0.868
5.2 V	4.550	4.544	228.158	0.006	1.162	1.390	5.09	0.836
5.5 V	4.834	4.830	381.200	0.004	1.101	1.478	2.89	0.746
6 V	5.315	5.312	626.110	0.003	0.999	1.625	1.60	0.615
7 V	6.294	6.292	1134.750	0.002	0.790	1.925	0.70	0.411
8 V	7.282	7.280	1647.250	0.002	0.580	2.227	0.35	0.260

5) Action 3: β versus V_B and α versus V_B

Plot of β versus V_B

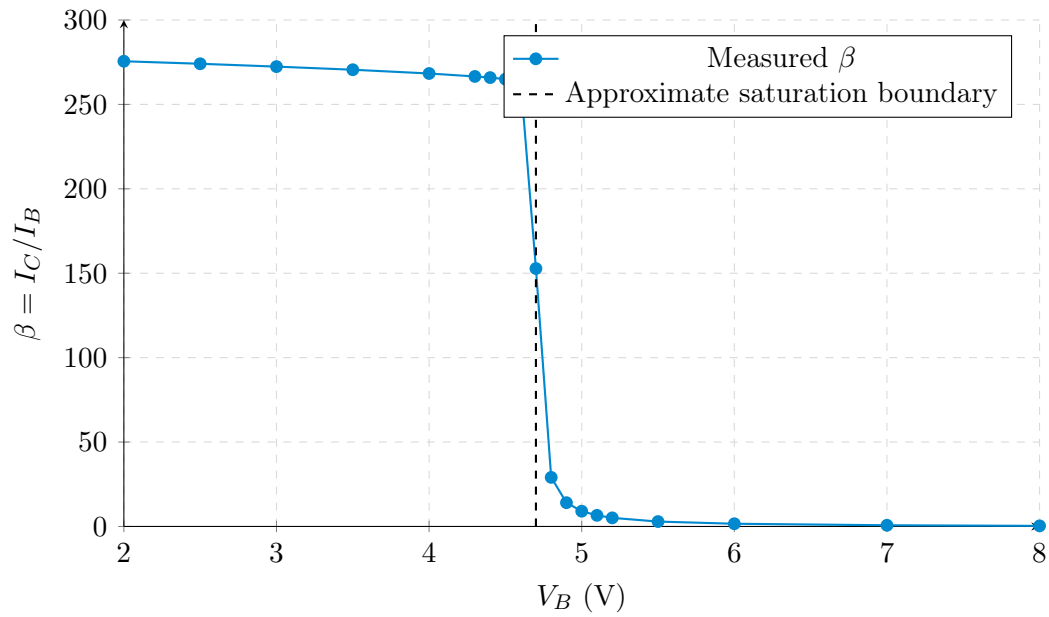


Figure 1: Common-emitter current gain β versus base voltage V_B

Plot of α versus V_B

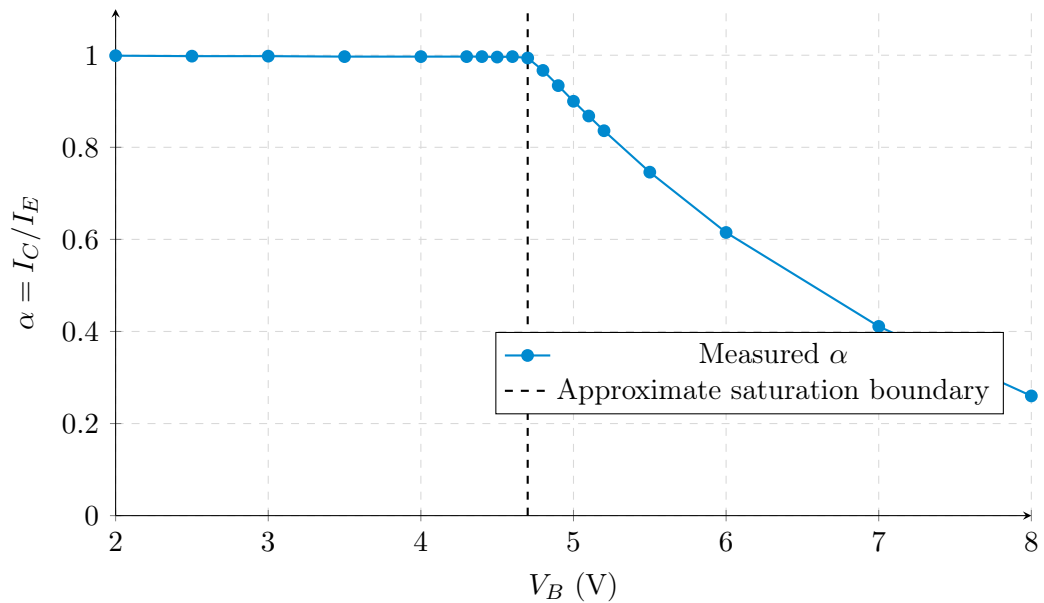


Figure 2: Common-base current gain α versus base voltage V_B

Answer

At low and moderate V_B , the BJT operates in the forward-active region. In this region, I_C is mainly controlled by I_B , so β stays large and relatively stable. From $V_B = 2$ V to

about 4.6 V, β remains around 264–276.

When V_B becomes higher, V_{CE} becomes very small and the transistor enters saturation. In saturation, I_C is limited by R_C and V_{CC} , while I_B keeps increasing. Therefore,

$$\beta = \frac{I_C}{I_B}$$

drops sharply.

The value of α also decreases at high V_B . In the forward-active region, α is close to 1. In saturation, more emitter current goes into the base, so

$$\alpha = \frac{I_C}{I_E}$$

becomes smaller.

6) Action 4: V_{CE} versus I_C

Plot of V_{CE} versus I_C

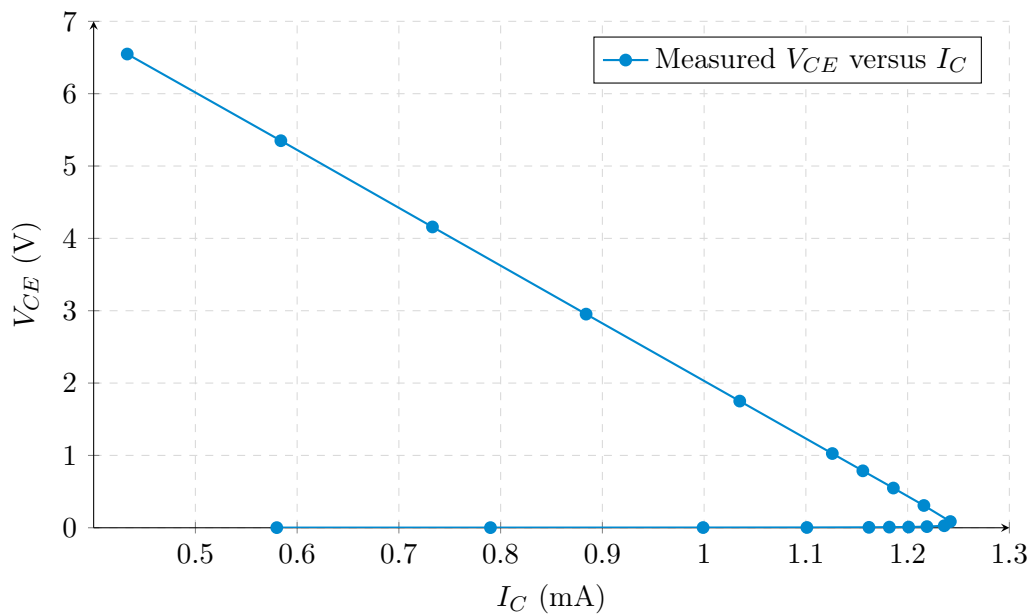


Figure 3: Collector-emitter voltage V_{CE} versus collector current I_C

Answer

The plot shows that V_{CE} decreases as I_C increases. This happens because a larger I_C causes a larger voltage drop across R_C , so V_C decreases. At the same time, V_E increases as V_B increases. Therefore,

$$V_{CE} = V_C - V_E$$

decreases quickly.

When V_{CE} becomes close to zero, the BJT enters saturation. In this region, I_C is no longer mainly controlled by I_B . Instead, it is limited by R_C and V_{CC} .

This plot is different from Lab 8 because Lab 8 measured the BJT output characteristic by changing V_{CE} while controlling I_B . In Lab 9, V_B is swept in a fixed circuit, so I_C and V_{CE} are connected by the load line:

$$V_C = V_{CC} - I_C R_C.$$

Thus, Lab 9 shows a load-line trajectory, while Lab 8 shows an output-characteristic curve.

7) Action 5: Required V_B for $V_C = V_{CC}/2$

Since

$$V_{CC} = 10.000 \text{ V}$$

the target collector voltage is

$$V_C = \frac{10.000}{2} = 5.000 \text{ V}$$

The measured base voltage required to make $V_C = V_{CC}/2$ was

$V_B = 5.67 \text{ V}$

This value is higher than the simple forward-active estimate because the transistor is already close to saturation around this operating condition. In saturation, the relationship between V_B and V_C is no longer described well by the simple approximation $V_E = V_B - 0.7 \text{ V}$ and $I_C \approx I_E$.

References

- [1] SCUPI 2026S Microelectronics Circuit Lab Manual, Lab #9: BJTs (2).
- [2] Sedra, A. S., Smith, K. C., Carusone, T. C., and Gaudet, V., *Microelectronic Circuits*, Oxford University Press.
- [3] Central Semiconductor Corp., *2N2222A NPN Silicon Transistor Datasheet*.